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Exploring Multi-Class Stress Detection Using Deep Neural Networks

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***Abstract*—This study uses Heart rate variability (HRV) as a biomarker to analyze the complex link between ongoing stress and its possible physical effects. To deal with difficult situations, people naturally feel stressed but when it lasts for a long time, stress can very badly affect the mental health. Depression, nervousness, and trouble sleeping are all common illnesses that can be caused by stress. Several advanced machine learning methods are used in this study to create a multi-class stress recognition model. The proposed stress detection model has achieved high accuracy, reaching 96.87% in stress identification using the SWELL-KW dataset. Leveraging this dataset, the study explores HRV traits as potential stress biomarkers, employing advanced machine learning methods including 1D CNN, 3D CNN, LSTM, GRU, and combinations such as LSTM + GRU + RNN. The main aim of this research work is to enhance the accuracy and utility of stress detection methods by investigating HRV traits as possible stress biomarkers. A key part of knowing how human bodies react to stress is being able to tell the difference between heart rate and HRV, especially when looking at the changes in RR intervals. Finally, the results of this study could help us learn more about how to recognize stress and come up with better ways to treat mental health problems that are caused by stress.**

Keywords - Heart rate variability (HRV), Stress biomarkers, Depression, Nervousness, Trouble sleeping, Multi-Class Stress Recognition model, RR intervals.

I. INTRODUCTION

In the quest to understand and address the intricate relationship between ongoing stress and its potential impact on our physical well-being, this study delves into the significance of HRV, a biomarker for stress. While stress is a natural response to challenging situations, its prolonged presence can significantly deteriorate mental health, giving rise to conditions such as depression, nervousness, and sleep disturbances. Recognizing the paramount importance of accurate stress detection, the study employs cutting-edge machine learning methods to construct a multi-class stress recognition model.

The focal point of the investigation lies in refining the precision and applicability of stress identification methodologies by scrutinizing HRV traits as potential stress biomarkers. Various advanced machine learning techniques, including 1D CNN, 3D CNN, LSTM, GRU, and combinations such as LSTM + GRU + RNN, are harnessed to enhance the efficacy of stress recognition models. A pivotal aspect of this exploration involves discerning the intricate interplay between heart rate and HRV, particularly in the context of analyzing changes in RR intervals.

The ultimate goal is to not only advance the accuracy of stress detection using HRV but also to glean insights into how our bodies uniquely respond to stressors. The anticipated findings hold promise in contributing to a nuanced understanding of stress recognition and, consequently, in devising more effective strategies for mitigating mental health problems induced by persistent worry. This study stands poised to deepen our comprehension of stress and pave the way for

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innovative approaches to alleviate stress-related mental health challenges.

II. LITERATURE SURVEY

This section examines the connection between HRV and stress by using a variety of techniques, such as physiological evaluations, smartphone apps, and machine learning methods. The studies offer insights into stress detection, its physiological characteristics, and potential applications in different domains.

HRV and Stress are thoroughly examined and evaluated in Kim et al.'s meta-analysis [1]. The study provides a comprehensive overview of the existing literature by highlighting the complex interplay between stress and HRV. Additionally, Muhajir et al. [2] provide an Android application that uses HRV analysis to assess stress levels. The work focuses on practical applications, demonstrating the potential of mobile technology in stress monitoring. Despite having a 70% accuracy rate, the classifier model may oversimplify the complex relationship between stress levels and HRV because it only uses linear analysis of HRV characteristics.

Several studies examine the effect of stress on HRV while performing cognitive tasks. Held et al. [3] explore HRV changes in individuals with anxiety during stressful cognitive tasks, shedding light on the physiological responses to stress in different populations. Furthermore, Dalmeida and Masala [4] explore HRV attributes as feasible physiological indicators for wearable device-based stress detection, highlighting the possibility of ongoing monitoring in real-world scenarios. Utilizing wearable sensors to record neuro-physiological signals may result in technical constraints and artefacts that affect the accuracy and dependability of the recorded data. A small sample size of 40 participants may limit statistical robustness. Ethical issues with consent and privacy with video data.

Miranda-Correa et al. [5] introduced the AMIGOS dataset, which is a valuable resource for research on mood, personality and affect states. The dataset contributes to understanding the link between emotional states and physiological signals, including HRV.

Self-evaluation and outside assessment techniques are used to annotate emotions; however, both approaches can be subjective and sensitive to individual biases, which may compromise the validity and reliability of the emotion labels applied to the data.

III. RELATED WORKS

A guideline for HRV measurement establishes standards for its measurement, physiological interpretation, and clinical use. [6] This standardization is crucial for ensuring consistency and reliability in HRV studies.

This research work also explores the potential of machine learning in stress detection. [7] Evaluating different machine learning strategies for heart rate-based stress detection with an emphasis on the importance of model setup emphasizes the importance of model configuration. [8] The multimodal machine learning and privacy-preserving compression models showcase the advancements in stress detection methodologies. [9] critically assess the accuracy of acquiring HRV from portable devices, highlighting the importance of reliability in wearable technology for stress monitoring. [10] emphasize the integration of wearable and flexible sensors for stress monitoring, introducing a novel approach to physiological data collection.

Machine learning's role in mental health showcases the versatility of these techniques in diverse domains. One of the research works [11] acknowledges the present diverse applications of machine learning in stress detection using physiological signals and behavioral patterns.

These research contributions highlight the multidisciplinary approach to stress detection by incorporating physiological measurements, mobile applications, and machine learning techniques. The integration of wearable technology and advanced analytics paves way for developing innovative solutions in stress monitoring, offering potential benefits for both research and practical applications. A crucial aspect of this study involves understanding the complex relationship between heart rate and HRV, particularly in the analysis of changes in RR intervals.

IV. PROPOSED WORK

The proposed framework leverages advanced machine learning techniques, including 1D CNN, 3D CNN, LSTM, GRU, and combinations like LSTM + GRU + RNN, to develop a robust multi-class stress recognition model. By converting the original variables into a new collection of uncorrelated variables called principal components, PCA is very helpful in finding the most informative characteristics and lowering the computing complexity of subsequent analyses. This is especially true when dealing with high-dimensional data. The arrangement of these principle components make it easier to pick and extract features, increasing the efficacy and efficiency of classification tasks. The first few components in the sequence capture the majority of the variance in the data. While accuracy is a widely used statistic, it is only adequate when the dataset is symmetric and the

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rates of false positives and false negatives are substantially comparable.
By analyzing HRV traits, this study seeks to enhance the precision and utility of stress identification methods.

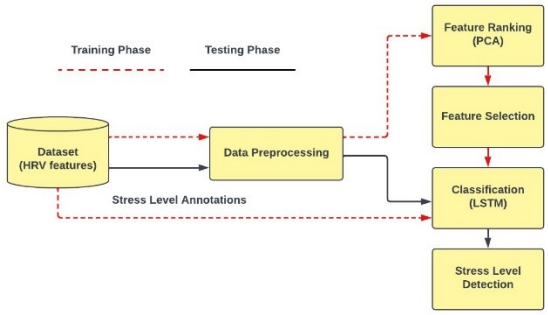

Fig 1: Framework of the proposed System

The proposed stress status classification model's framework is shown in Fig.1; it goes from the collection of data to stress level classification. The ultimate goal is to contribute valuable insights into understanding stress-related physical effects, paving the way for improved recognition and treatment of mental health issues stemming from prolonged stress.

A. Dataset Collection

The SWELL-KW dataset from Kaggle was used for developing the proposed stress recognition system is carefully curated to capture diverse physiological responses to stress, with a specific focus on heart rate variability (HRV) as the primary biomarker. The dataset was partitioned into training and testing subsets using an 80/20 split, where 80% of the data was allocated to the training set and the remaining 20% was reserved for testing.

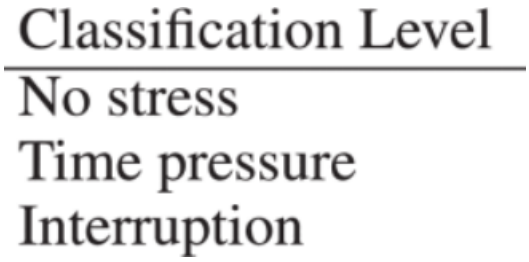

Fig 2: Stress Classes in SWELL-KW

The distribution of the collected data with three different Stress classes are presented in Fig. 2. The dataset encompasses a wide range of individuals across different demographic groups, ensuring a representative sample for robust model training and evaluation.

Table 1. Explanation

No	*Feature*	*Meaning*
1	MEAN_RR	Mean of RR intervals
2	MEDIAN_RR	Median of RR intervals
3	LF_NU	Low frequency activity
4	VLF	Very low frequency activity
5	TP	Total HRV power spectrum
6	SDRR_RMSSD_REL_RR	Ratio of SDRR_REL over RMSSD_REL
7	LF_HF	Ratio of low to high frequency
8	SDSD	SD of successive RR interval differences
9	SD1	Measures short-term HRV in ms and correlates with BRS
10	RMSSD	Root mean square of successive RR interval differences
11	Sampen	Sample entropy of the RR sign
12	pNN25	Percentage of successive RR intervals that differ more than 25 ms
13	VLF_PCT	Very low frequency activity of the HRV Spectrum
14	SDRR	SD of RR intervals
15	SD2	Measures long-term HRV in ms and correlates with (BRS)

Demographic Information:
Age: The dataset includes participants spanning various age groups to account for age-related variations in stress responses.
Gender: Both male and female participants are represented to ensure gender diversity in stress recognition models.
Ethnicity: The dataset considers diverse ethnic backgrounds to address potential variations in stress responses among different populations.

B. Pre-processing

To perform data analysis and visualization, the initial step involves importing essential libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Once these libraries are imported, the next step is to load the HRV dataset using Pandas, typically achieved through the pd.read_csv() function. To conduct Exploratory Data Analysis (EDA), it is crucial to examine null values in the dataset using df.isnull().sum() to identify and handle missing data appropriately. Visualization is integral for understanding the dataset, and libraries like Matplotlib and Seaborn can be employed for creating insightful plots, histograms, and heatmaps. For feature extraction, the scikit-learn library allows for the implementation of Principal Component Analysis (PCA). The implementation of PCA facilitates the reduction of dataset dimensionality and the identification of the most important features. After PCA is applied, the top 15 PCA sorted features are

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shown in Table 1, selected for training and testing purposes Normalization through scaling, a crucial step for enhancing model performance, can be accomplished using preprocessing techniques like Min-Max scaling or Standard Scaling provided by scikit-learn's MinMaxScaler or StandardScaler classes. In summary, the data processing involves importing libraries, loading the dataset, examining null values, performing visualization for EDA, implementing PCA for feature extraction, and normalizing data through scaling for optimal model performance.

C. Training & Testing

To implement the described modules for a predictive model, the process involves several key steps. First, the "train_test_split" module from a machine learning package such as scikit-learn must first be used to split the data into training and testing sets. By using this step, it is ensured that the model is assessed on an unseen component of the data and trained on a fraction that has been seen. Next, model generation include building different kinds of neural network designs, like 1D CNN, 3D CNN, LSTM, GRU, and combinations like LSTM + GRU + RNN. Libraries like TensorFlow or PyTorch can be utilized for this purpose. Each architecture may be suitable for different types of data or specific tasks. Finally, the prediction module processes the user input through the trained model and displays the final predicted output. This can be achieved by integrating the trained model into the application and utilizing the input from the user interface. In summary, the implementation involves data splitting, model generation, user signup/login, user input collection, and prediction, with each module seamlessly integrated to create a comprehensive predictive application.

D. Algorithms

A particular kind of neural network called a 1D CNN is intended to process data in one-dimensional sequences such as time series or sequential data. It employs convolutional layers to capture local patterns and hierarchical features within the input sequence.
A 3D CNN extends the concept of convolutional layers into three dimensions, making it particularly suitable for processing volumetric data like video sequences or medical imaging. This architecture considers the spatial and temporal dimensions simultaneously, allowing it to capture spatiotemporal features.
A specific kind of recurrent neural network (RNN) architecture called LSTM is meant to solve the problem of vanishing gradients that standard RNNs have. Because memory cells and gates are integrated into LSTMs, they excel at capturing intricate dependencies within sequential data.
GRU is another form of recurrent neural network aimed to get over the drawbacks of traditional RNNs, much like LSTM. By combining the hidden state and memory cell, GRUs simplify the design and reduce training time and parameter requirements. While
LSTM + GRU + RNN:
This hybrid approach aims to efficiently capture dependencies of both short-term and long-term in sequential data. By incorporating LSTM for long-term memory, GRU for computational efficiency, and RNN for simplicity, this combined architecture can achieve better performance in various applications, balancing the trade-offs among memory, computation, and training speed.

V. RESULTS AND DISCUSSION

The experiment's outcomes show how useful biomarkers and cutting-edge advanced machine learning methods are for precisely identifying stress. The model achieves great accuracy in identifying subtle stress levels based on HRV features by combining 1D CNN, 3D CNN, LSTM, and GRU. These discoveries deepen our knowledge of the physiological consequences of stress, which will help develop better therapies and insightful new perspectives on the advancement of mental health services.

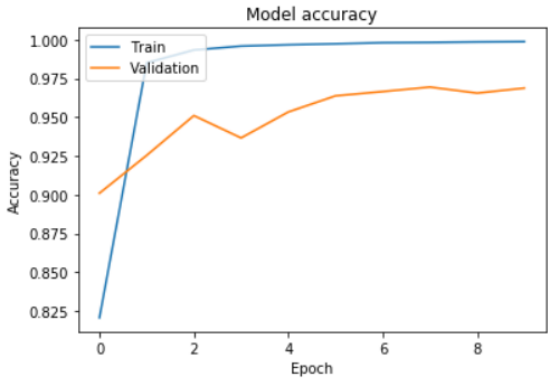

Fig 3: Training versus Validation accuracy of LSTM model

Overall, LSTM achieved high accuracy in stress identification when utilizing advanced machine learning methods. Additionally, we have confirmed if the suggested model is overfitted or not. The training versus validation accuracy found in our experiments is shown in Fig. 3. This chart shows that the validation loss is marginally more than the training loss, and that the validation accuracy and training accuracy are

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almost equal. Stated differently, the model satisfies the requirements for a decent fit and is not overfitted. By using only, the top 15 PCA-sorted features, we further examine the model's performance. The findings are shown in Table. 2.

Table 2. Performance of the proposed LSTM model for three level classifications with top 15 PCA-sorted features.

Classification level	***Precision***	***Recall***	***F1-Score***
No stress	0.97	0.98	0.98
Time Pressure	0.97	0.95	0.96
Interruption	0.95	0.96	0.95

Accuracy: A test's accuracy is defined as its ability to recognize debilitated and solid examples precisely. To quantify a test's exactness, we should register the negligible part of genuine positive and genuine adverse outcomes in completely examined cases.
Accuracy = (TP + TN) / (TP + TN + FP + FN)

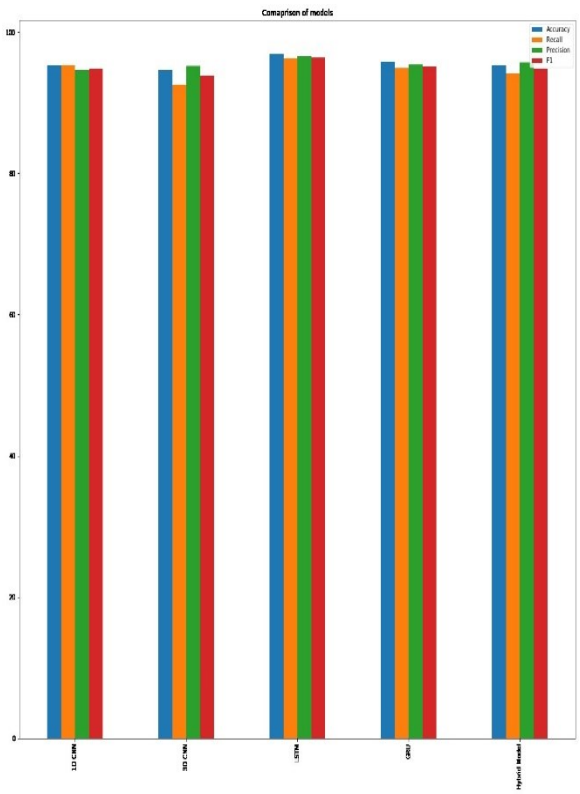

Fig 4: Comparison of various models using Accuracy, Recall, Precision, F1-Score

Precision: The fraction of accurately categorized positive samples or events among all positive predictions is known as precision.

Precision = TP / (TP + FP)

Recall: Recall is a machine learning metric that surveys a model's capacity to recognize all pertinent examples of a particular class. It represents the ratio of correctly predicted positive instances to add up to real up-sides, which gives data about a model's capacity to catch instances of a specific class.

Recall = TP / (TP + FN)

F1-Score: The F1 score is a machine learning evaluation measurement that evaluates the precision of a model. It consolidates a model's precision and review scores. The precision measurement computes how often a model anticipated accurately over the full dataset.
F1-score = (2 × Recall × Precision) / (Recall + Precision)

Various neural network architectures were evaluated for stress detection using HRV features, considering metrics such as accuracy, recall, precision, and F1 score, as depicted in Fig 4.

Table 3. Performance Evaluation of different models

Model	**Accuracy**	**Recall**	**Precision**	**F1-Score**
1D CNN	95.3086	95.2601	94.5742	94.8427
3D CNN	94.6384	92.5832	95.2430	93.8065
LSTM	96.8732	96.3125	96.5318	96.4190
GRU	95.8009	94.9236	95.3978	95.1073
Hybrid Model	95.3062	94.0960	95.6789	94.8491

As shown in Table 3, The 1D CNN model achieved a high accuracy of 95.31%, with strong performance in recall (95.26%) and precision (94.57%), resulting in an F1 score of 94.84%. The 3D CNN model demonstrated a slightly lower accuracy of 94.64%, with a balanced performance in recall (92.58%) and precision (95.24%), resulting in an F1 score of 93.81%. The LSTM model outperformed other architectures with the highest accuracy of 96.87%, along with excellent recall (96.31%) and precision (96.53%), leading to an impressive F1 score of 96.42%. The GRU model achieved a commendable accuracy of 95.80%, with balanced recall (94.92%) and precision (95.40%), resulting in an F1 score of 95.11%. The Hybrid Model, which presumably combines features from multiple architectures, achieved a competitive accuracy of 95.31%, with relatively lower recall (94.10%) but higher precision (95.68%), resulting in an F1 score of 94.85%.

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VI. CONCLUSION

In conclusion, this study leverages advanced machine learning techniques, including 1D CNN, 3D CNN, LSTM, GRU, and combinations like LSTM + GRU + RNN, to construct a robust multi-class stress recognition model utilizing HRV as a biomarker. The most efficient neural network architecture for stress detection using HRV characteristics is the LSTM model, which outperformed the other models in terms of accuracy, recall, precision, and F1 score. While the 3D CNN and hybrid models had somewhat lower but still acceptable performance in stress detection, the 1D CNN and GRU models also demonstrated outstanding performance. The investigation sheds light on the intricate relationship between persistent stress and its potential physiological impacts, emphasizing the critical need to accurately identify stress for effective mental health management. Recognizing that stress is a natural response to challenging situations, the study underscores the detrimental effects of prolonged stress on mental well-being, contributing to prevalent issues such as depression, nervousness, and sleep disturbances. By delving into HRV traits as potential stress biomarkers, the research aims to enhance the precision and practicality of stress detection methods. The diverse array of machine learning methods employed in this study demonstrates a comprehensive approach to stress recognition, offering promising avenues for refining our understanding of stress and devising improved strategies for addressing mental health issues stemming from stress-induced concerns. Ultimately, the findings hold the potential to advance the field, providing valuable insights into stress recognition and paving the way for more effective treatments for stress-related mental health problems.

VII. FUTURE WORK

The future scope of this research lies in the continued refinement and real-world implementation of the developed stress recognition model. Further exploration could involve incorporating real-time data streams for dynamic stress monitoring, optimizing model interpretability, and adapting the framework for personalized stress management interventions. Collaborations with healthcare professionals and technology developers may facilitate the integration of this model into wearable devices, enabling widespread accessibility. Additionally, exploring the applicability of the model across diverse demographic groups and cultural contexts could enhance its universal effectiveness. Continual updates and validation through longitudinal studies will be essential for ensuring the model's relevance and reliability in evolving mental health landscapes.

REFERENCES

[1] H.-G. Kim, E.-J. Cheon, D.-S. Bai, Y. H. Lee, and B.- H. Koo, ‘‘Stress and heart rate variability: A meta-analysis and review of the literature,’’ Psychiatry Invest., vol. 15, no. 3, pp. 235–245, Mar. 2018.

[2] D. Muhajir, F. Mahananto, and N. A. Sani, ‘‘Stress level measurements using heart rate variability analysis on Android based application,’’ Proc. Comput. Sci., vol. 197, pp. 189–197, Jan. 2022.

[3] J. Held, A. Vîslă, C. Wolfer, N. Messerli-Bürgy, and C. Flückiger, ‘‘Heart rate variability change during a stressful cognitive task in individuals with anxiety and control participants,’’ BMC Psychol., vol. 9, no. 1, p. 44, Mar. 2021.

[4] K. M. Dalmeida and G. L. Masala, ‘‘HRV features as viable physiological markers for stress detection using wearable devices,’’ Sensors, vol. 21, no. 8, p. 2873, Apr. 2021.

[5] J. A. Miranda-Correa, M. K. Abadi, N. Sebe, and I. Patras, ‘‘AMIGOS: A dataset for affect, personality and mood research on individuals and groups,’’ IEEE Trans. Affect. Comput., vol. 12, no. 2, pp. 479–493, Apr./Jun. 2021.

[6] M. Malik, J. T. Bigger, A. J. Camm, R. E. Kleiger, A. Malliani, A. J. Moss, and P. J. Schwartz, ‘‘Heart rate variability: Standards of measurement, physiological interpretation, and clinical use,’’ Eur. Heart J., vol. 17, no. 3, pp. 354–381, Mar. 1996.

[7] M. Albaladejo-González, J. A. Ruipérez-Valiente, and F. G. Mármol, ‘‘Evaluating different configurations of machine learning models and their transfer learning capabilities for stress detection using heart rate,’’ J. Ambient Intell. Human. Comput., pp. 1–11, Aug. 2022, doi: 10.1007/s12652- 022-04365-z.

[8] A.Ibaida, A. Abuadbba, and N. Chilamkurti, ‘‘Privacy-preserving compression model for efficient IoMT ECG sharing,’’ Comput. Commun., vol. 166, pp. 1–8, Jan. 2021.

[9] W. C. Dobbs, M. V. Fedewa, H. V. MacDonald, C. J. Holmes, Z. S. Cicone, D. J. Plews, and M. R. Esco, ‘‘The accuracy of acquiring heart rate variability from portable devices: A systematic review and meta-analysis,’’ Sports Med., vol. 49, no. 3, pp. 417–435, Mar. 2019

[10] M. X. Huang, J. Li, G. Ngai, and H. V. Leong, ‘‘Stress clicks: Sensing stress from gaze-click patterns,’’ in Proc. 24th ACM Int. Conf. Multimedia (MM), Oct. 2016, pp. 1395–1404

[11] B. J. Feir-Walsh and L. E. Toothaker, ‘‘An empirical comparison of the ANOVA F-test, normal scores test and Kruskal–Walli’s test under violation of assumptions,’’ Educ. Psychol. Meas., vol. 34, no. 4, pp. 789–799, Dec. 1974.

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